Algebraic Geometry and Galois Theory

(= field theory)

via linear algebra.

$\dim_{\mathbb{F}} E < \infty$

- Finitely generated field extensions

"$(\mathbb{F}, +_{\mathbb{F}}, \times_{\mathbb{F}}) \subseteq (E, +_E, \times_E)$" $\Rightarrow E$ is a v.s/$\mathbb{F}$.

$+_E|_{\mathbb{F}}$ $\quad$ $\times_E|_{\mathbb{F}}$

$+_E$

- $\cdot : \mathbb{F} \times E \to E$

  $a \cdot u \mapsto au$

$+_E : E \times E \longrightarrow E$

$\times_E : E \times E \longrightarrow E$

$\cdot = \times_E|_{\mathbb{F}\times E} : \mathbb{F} \times E \longrightarrow E$

$\rightsquigarrow +_E|_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \longrightarrow \mathbb{F}$

$+_{\mathbb{F}}$ $\quad$ $(a, b) \longmapsto a +_E b$

$\times_E|_{\mathbb{F}} : \mathbb{F} \times \mathbb{F} \longrightarrow \mathbb{F}$

$\times_{\mathbb{F}}$

• Function fields of Varities.

a space • Top. sp $(X, \mathcal{T})$

• Smooth manifold.

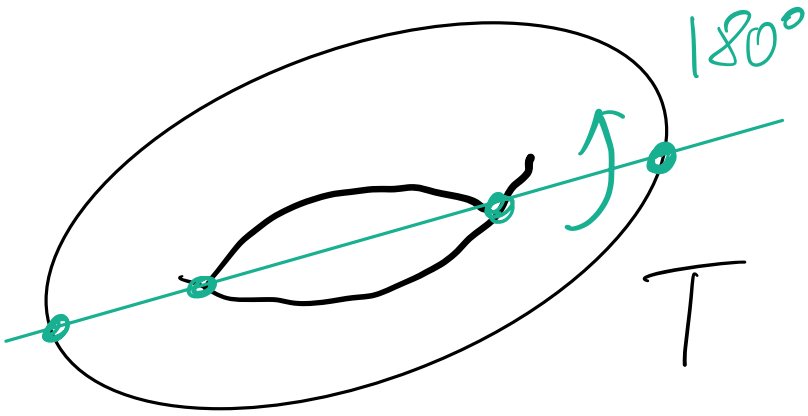
$\forall p \in M$, "$\exists$ nbhd

U_p of p s.t

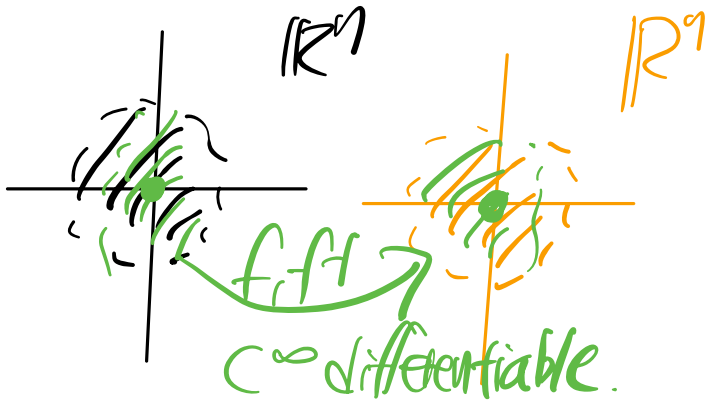
$U_p \cong \mathbb{B}^n_R$.

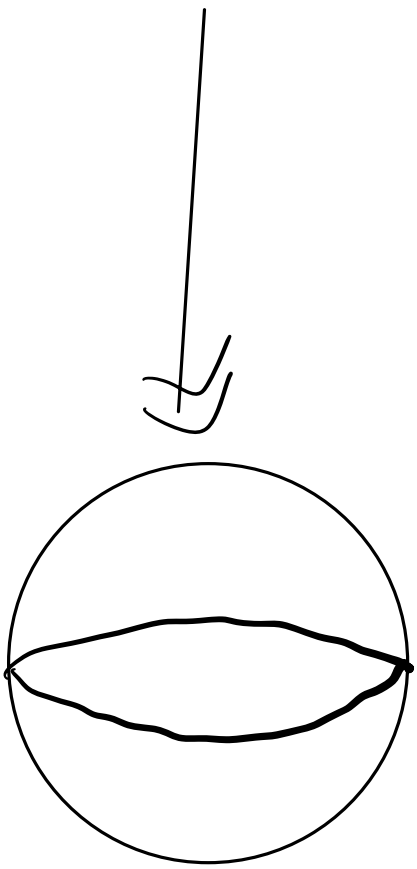
$\exists$ homeo

Illustrative Example









$T/\sim \cong S^2$ • Variety.

= Zeroset of polynomial locally.

Realization of smooth manifolds as varieties

In general, this fails.

This happens for • Compact Riemann surfaces

• projective manifold over $\mathbb{C}$.

$\Leftrightarrow$ a closed submanifold of $\mathbb{P}^N_{\mathbb{C}}$ for some $N \in \mathbb{N}$.

$M \overset{\varphi}{\hookrightarrow} \mathbb{P}^N_{\mathbb{C}}$

; φ is 1-1. } embedding

; $d\varphi$ is 1-1 }

$\Rightarrow M \overset{\varphi}{\cong} \mathrm{Im}\,\varphi \subseteq \mathbb{P}^N_{\mathbb{C}}$ ← closed set.

due to Chow's theorem.

k a field.

$$\mathbb{P}^N_k := k^{N+1} - \{(0,\dots,0)\} \Big/ x \sim y$$

$\ni [X_0 : \dots : X_N]$

$\overset{\text{def}}{=} [\lambda X_0 : \dots : \lambda X_N]$ for $\lambda \in k \setminus \{0\} = k^*$.

$k = \mathbb{C}$, $\mathbb{Z}_2 = \{0,1\}$, $\mathbb{Q}$, k: number field.

- $S^2 = \mathbb{P}^1_{\mathbb{C}} \xhookrightarrow{\varphi} \mathbb{P}^2_{\mathbb{C}}$

$$[x:y] \longmapsto [x^2 : xy : y^2]$$

$$X = Y = Z, \quad XZ = Y^2.$$

Exercise Show that φ, $d\varphi$ are injective.

$$\Rightarrow \mathbb{P}^1_{\mathbb{C}} \cong \operatorname{Im}\varphi = Z(XZ - Y^2)$$

$$= \{[X:Y:Z] \in \mathbb{P}^2_{\mathbb{C}} \mid XZ = Y^2\}$$

$$S^2 = \mathbb{P}^1_{\mathbb{C}} \xhookrightarrow{\varphi_3} \mathbb{P}^3_{\mathbb{C}}$$

$$[x:y] \longmapsto [x^3 : x^2y : xy^2 : y^3]$$



$$Y^2 = XZ.$$
$$Z^2 = YW.$$

$$\mathbb{P}^1_{\mathbb{C}} \cong \mathrm{Im}\,\varphi_3$$

$$= \{[X:Y:Z:W] \in \mathbb{P}^3_{\mathbb{C}} \mid YZ = XW,\ Y^2 = XZ,\ Z^2 = YW\}.$$

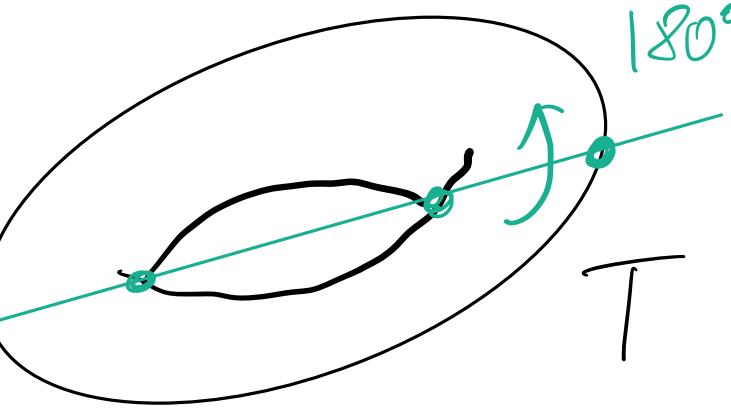
- $\mathbb{P}^1 \ni [x:y] \overset{x\neq 0}{=} [1:\frac{y}{x}]$
  $\overset{y\neq 0}{=} [\frac{x}{y}:1]$.

- Torus $\xhookrightarrow{\varphi}$ $\mathbb{P}^2_{\mathbb{C}}$.

$$\mathrm{Im}\,\varphi = Z(Y^2Z = X^3 + aXZ^2 + bZ^3)$$

in $\mathbb{P}^2_{\mathbb{C}}$.

$X \mapsto \lambda X$
$Y \mapsto \lambda Y$
$Z \mapsto \lambda Z$.



T

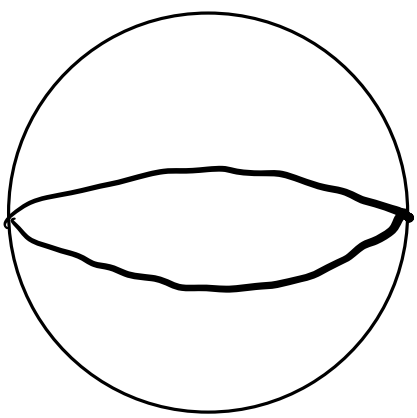
$V(Y^2Z - X^3 - aXZ^2 - bZ^3)$

$\lambda^3($ $)$ elliptic curve

$\downarrow$ ufl

$\leftrightarrow$ variety

$\downarrow$



$T/\sim$

$V(XZ - Y^2)$ in $\mathbb{P}^2_{\mathbb{C}}$.

rational curve.

Choose $Z \neq 0$ for simplicity

($\mathbb{C}$ inhomogeneous open set)

$$y = \frac{Y}{Z}, \quad x = \frac{X}{Z}.$$

$$\lambda = \frac{1}{Z}$$

$$[X:Y:Z] = \left[\frac{X}{Z}:\frac{Y}{Z}:1\right]$$

$$= [x:y:1]$$

$$= (x,y) \in \mathbb{C}^2.$$

E

$\|$



$\|$

$$V(Y^2Z - X^3 - aXZ^2 - bZ^3) \quad Z = 1.$$

$$\overset{Z \neq 0}{=} \{(x,y) \in \mathbb{C}^2 \mid y^2 = x^3 + ax + b\}$$

$V(XZ-Y^2) \cong \mathbb{CP}^1 = \{[X:Y]\}$

Choose $Y \neq 0$. $x = \frac{X}{Y} \in \mathbb{C}^1$.

$$E \xrightarrow{2:1 \text{ map}} \mathbb{P}^1_{\mathbb{C}}$$

$$(x,y) \longmapsto x$$

$y^2 = x^3 + ax + b$

$y^2 = \square$, $y = \sqrt{\square}, -\sqrt{\square}$.

$$Q(\mathbb{P}^1_{\mathbb{C}}) = \mathbb{C}(x) = F.$$

$$Q(E) = \mathbb{C}(x)(y) / (y^2 - x^3 - ax + b).$$

$$= E.$$

$$F \subseteq E.$$

$$\mathrm{Gal}(E/F) = \mathbb{Z}_2.$$